

MEDIATEK

MT8676 Yocto Display User Manual

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Version History

Version	Date	Author	Description
1.0	2024-08-12	Scott Wang	Official release

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1 Display & Multi-Screen

1.1 Overview

1.1.1 Brief Introduction

This chapter mainly introduces the MT8676 display architecture and multi-screen configuration guide.

1.1.2 Abbreviation

Table 1-1. Abbreviation

Abbreviation	Explanation
CRTC	Cathode Ray Tube Controller
DP	DisplayPort
DPI	Digital Parallel Interface
DSC	Display Stream Compression
DSI	Display Serial Interface
DTS	Device Tree
HWC	Hardware Composer
LCM	Liquid Crystal Display Module
SerDes	Serializer/Deserializer

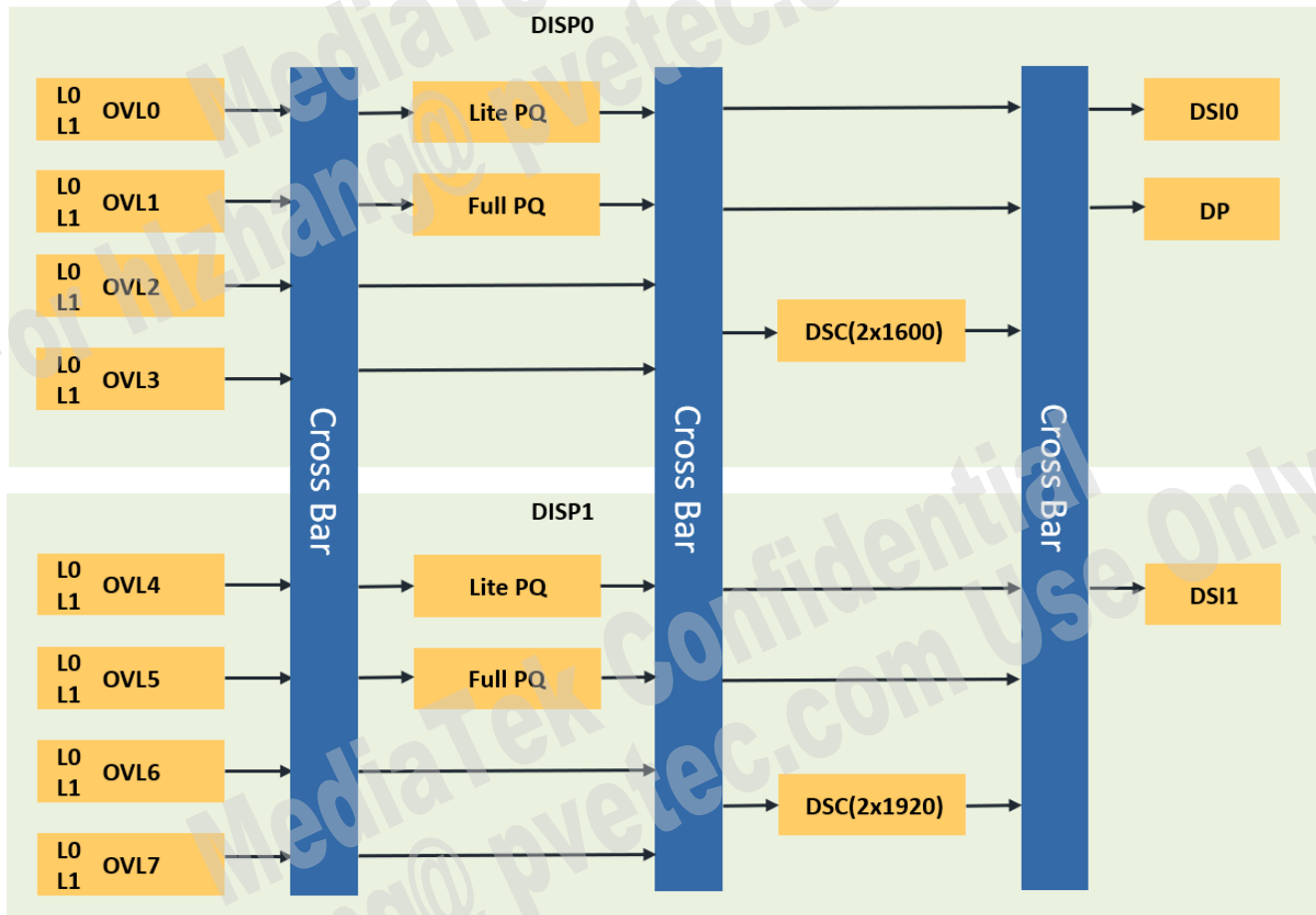
1.1.3 Capability

Table 1-2. Capability

DISP pipeline	10bit pipe x 3 (MAX 688MHz at 0.75V)
DSI	DSI0 + DSI1 C/D PHY Combo 4-lane x 2 DPHY: 2.5Gbps/lane
DP	DP1.4, 4-lane 8.1Gbps/lane (4-lane mode conflict with USB3)
Panel number	1~6
OVL layers	OVL0~OVL7: Total 16 layers
OVL De-comp.	AFBC (RGBA8888/RGB888/RGB565)
Compression	VESA DSC 1.2 (2 slice x 2)

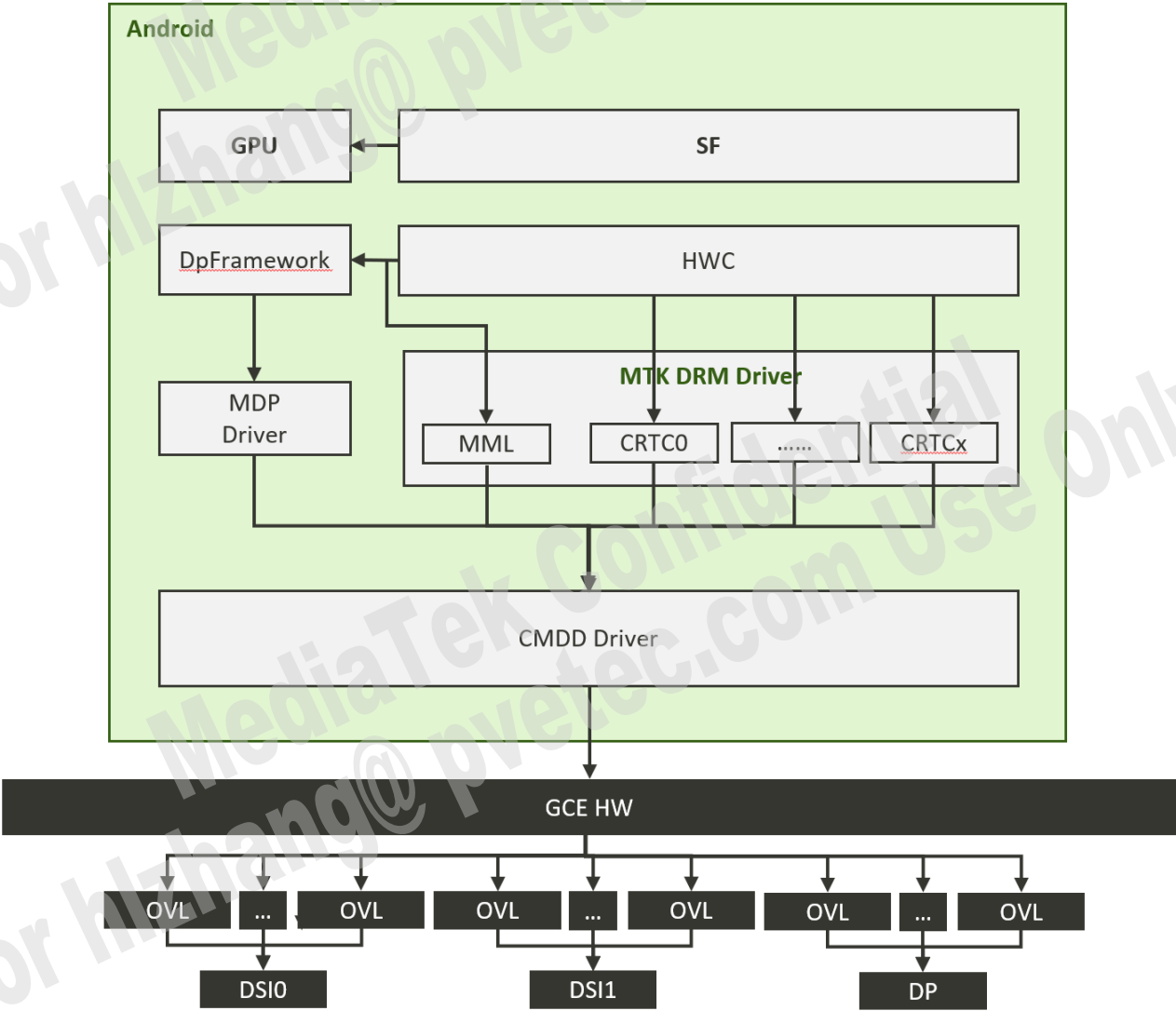
1.2 Architecture/Process Overview

1.2.1 HW Architecture

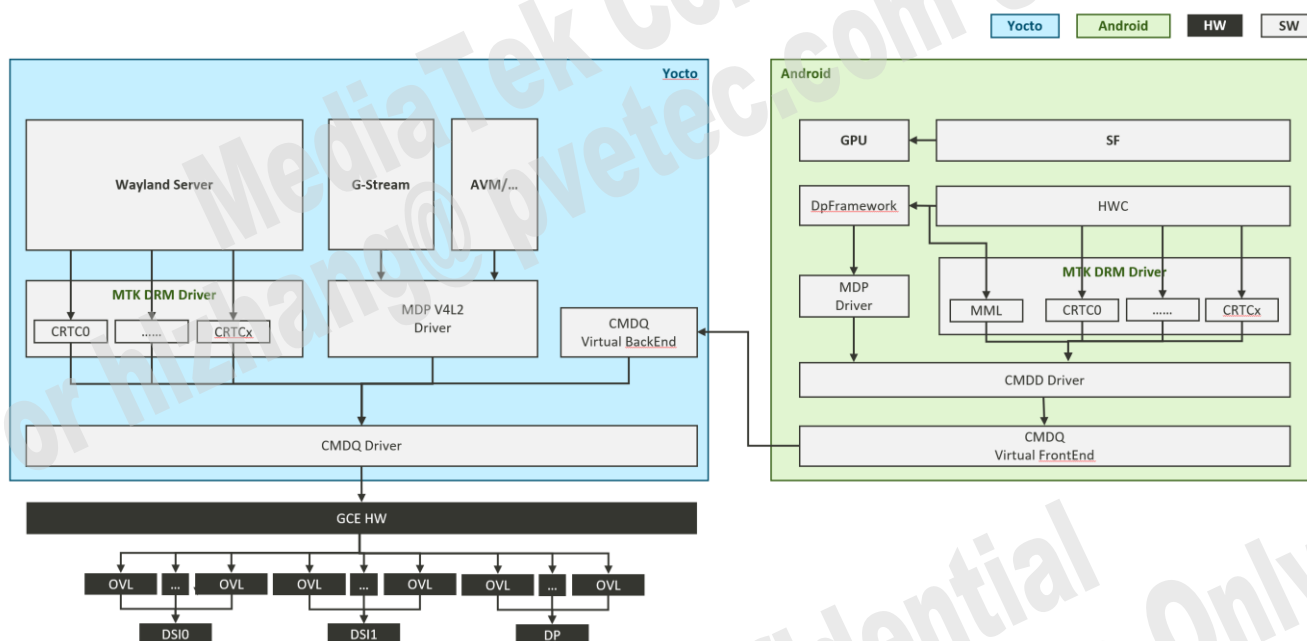


1.2.2 SW Architecture

1.2.2.1 Android



1.2.2.2 Hypervisor (Yocto + Android)



1.3 Configuration/Customization Guideline

1.3.1 DSI SuperFrame Dual Panel Configuration

DSI0/1 supports outputting side by side superframe and driving dual-screen display.

You need to open the virt_dsi node in dts (the figure below is a DSI0 configuration sample). For other LCM configurations, please refer to *MT8676_DSI_Panel_User_Manual_V1.0*.

```
&virt_dsi0_1 {
    status = "okay";
};
```

1.3.2 DP SuperFrame Dual Panel Configuration

DP supports outputting side by side superframe and driving dual-screen display.

You need to open the virt_dp node in dts. For other LCM configurations, please refer to *MT8676_DSI_Panel_User_Manual_V1.0*.

```
&virt_dp_1 {
    status = "okay";
};
```

1.3.3 LCM Rotation Configuration

Path: device/mediateksample/project/device-vext.mk

```
PRODUCT_PROPERTY_OVERRIDES += ro.vendor.sf.orientation_external1=xxx
```

Above `ro.vendor.sf.orientation_externalX` indicates the rotation angle of the corresponding X screen. If the screen is in portrait or landscape orientation, it generally needs to be set to 90°.

1.3.4 OVL Configuration

You can configure the OVL usage of each CRTC in the `dispsys_config` node in DTS. The following sample code configures four OVLs for CRTC0, for a total of eight layers.

```
&dispsys_config {
    crtc0-ovl-path = <DDP_COMPONENT_OVL0_2L>,
        <DDP_COMPONENT_OVL1_2L>,
        <DDP_COMPONENT_OVL2_2L>,
        <DDP_COMPONENT_OVL3_2L>,
        <DDP_COMPONENT_OVLSYS_DLO_ASYNC3>;
    crtc-layer-nr = <8>;
    pinctrl-names = "mode_te_te";
    pinctrl-0 = <&mtkfb_pins_lcm_dsi_te>;
    status = "okay";
};
```

1.4 Frequently Asked Questions/Troubleshooting

1.4.1 hwc CMD

Log cmd:

```
adb shell setprop persist.vendor.debug.hwc.log V && adb shell setprop
vendor.debug.hwc.skip_log 0 && adb shell dumpsys SurfaceFlinger
```

Dump sf info:

```
adb shell dumpsys SurfaceFlinger > sf.log
```

Force gpu (gpu compose)

```
adb shell service call SurfaceFlinger 1008 i32 1
```

1.4.2 Display Driver CMD

On means enable log output, Off means disable log output.

Frequently used display mobile log:

```
adb shell "echo mobile:on > /d/mtkfb"
```

If you want to capture the boot log, you need to modify the code directly:

```
kernel/kernel_device_modules-6.1/drivers/gpu/drm/mediatek/mediatek_v2/mtk_debug.c
```



```
bool g_mobile_log = 1;
```

More detail log:

```
adb shell "echo detail:on > /d/mtkfb"
```

If you want to capture the boot log, you need to modify the code directly:

```
kernel/kernel_device_modules-6.1/drivers/gpu/drm/mediatek/mediatek_v2/mtk_debug.c
```

```
bool g_detail_log = 1;
```

Fence log:

```
adb shell "echo fence:on > /d/mtkfb"
```

If you want to capture the boot log, you need to modify the code directly:

```
kernel/kernel_device_modules-6.1/drivers/gpu/drm/mediatek/mediatek_v2/mtk_debug.c
```

```
bool g_fence_log = 1;
```

Irq log:

```
adb shell "echo irq:on > /d/mtkfb"
```

If you want to capture the boot log, you need to modify the code directly:

```
kernel/kernel_device_modules-6.1/drivers/gpu/drm/mediatek/mediatek_v2/mtk_debug.c
```

```
bool g_irq_log = 1;
```

Get display diagnose dump log:

```
adb shell "echo diagnose > /sys/kernel/debug/mtkfb && cat /sys/kernel/debug/mtkfb" > mtkfb.txt
```

1.4.3 Pattern

First, you need to open the register debug.

```
diff --git a/drivers/clk/mediatek/clkdbg.c b/drivers/clk/mediatek/clkdbg.c
index 124dff7d05c1..960b294e6cf5 100644
--- a/drivers/clk/mediatek/clkdbg.c
+++ b/drivers/clk/mediatek/clkdbg.c
@@ -980,7 +980,7 @@ static int clkdbg_set_rate(struct seq_file *s, void *v)
     return r;
 }

-#if defined(CONFIG_MTK_ENG_BUILD)
+#if 1//defined(CONFIG_MTK_ENG_BUILD)
 static void *reg_from_str(const char *str)
 {
     static phys_addr_t phys;
@@ -2164,7 +2164,7 @@ static const struct cmd_fn common_cmds[] = {
     CMDFN("disable_unprepare_provider", clkdbg_disable_unprepare_provider),
     CMDFN("set_parent", clkdbg_set_parent),
     CMDFN("set_rate", clkdbg_set_rate),
-#if defined(CONFIG_MTK_ENG_BUILD)
+#if 1//defined(CONFIG_MTK_ENG_BUILD)
     CMDFN("reg_read", clkdbg_reg_read),
     CMDFN("reg_write", clkdbg_reg_write),
     CMDFN("reg_set", clkdbg_reg_set),
```

- dsi0 pattern:

```
adb shell "echo reg_write 0x1400d178 0xc61 > /proc/clkdbg;cat /proc/clkdbg"
```

- dsi1 pattern:

```
adb shell "echo reg_write 0x1420d178 0xc61 > /proc/clkdbg;cat /proc/clkdbg"
```

- dp pattern:

```
adb shell "echo reg_write 0x1400bf00 0x41 > /proc/clkdbg;cat /proc/clkdbg"
```

1.4.4 Black Screen

- Is the backlight on?
- If a bridge IC exists, is the bridge IC OK?
- Is the dsi/dp pattern OK?
- Is the screencap OK?

```
adb shell screencap -d 0/1 /sdcard/1.png
```

After confirming that all the above check points are satisfactory, the display owner should make analysis.

1.4.5 Screen Distortion/Flickering

- Check the log to see if there is DISP_OVL/RDMA underflow/abnormal

- Check size setting

- cmd: adb shell "echo mobile:on > /d/mtkfb"

- Check clk/dram frequency

- clk

- dump:

```
adb shell "echo fmeter > /proc/clkdbg ; cat /proc/clkdbg | grep -e disp"
```

- o force max:

```
adb shell "echo 0 0 > /sys/module/mtk_mmdvfs_debug/parameters/force_step"
```

- dvfs

- o dump:

```
adb shell "cat /sys/kernel/helio-dvfsrc/dvfsrc_dump | grep -e uv -e Mbps"
```

- o force max:

```
adb shell "echo 0 > /sys/kernel/helio-dvfsrc/dvfsrc_force_vcore_dvfs_opp"
```

- Is force GPU OK

```
adb shell service call SurfaceFlinger 1008 i32 1
```

- Is dsi/dp pattern OK

1.4.6 Screen Freeze/Lag

- Check main log to see if there is fence timeout

```
GED : Frame didn't finished in 1000 ms (try = 1)
GED : Fence fd = 82
GED : fence SurfaceView - #0:2 0
GED : pt timeline-primary-0-13 sw_sync status(0) sync_drv(250117) 0.000000
GED : pt timeline-unknown-0-13 sw_sync status(0) sync_drv(250048) 0.000000
GED : pt timeline-external-3-16 sw_sync status(1) sync_drv(238184) 8203.513127
```

- Check kernel log to see if there is underflow/abnormal
- Enable fence log, check fence release

```
adb shell "echo fence:on > /d/mtkfb"
```

- User systrace to analysis performance issue

```
adb shell perfetto -o /data/misc/perfetto-traces/trace -t 10s sched freq idle am wm gfx
view input
```

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